

The long-term effects of the free primary education policy on intergenerational mobility in Kenya: A household survey in 2015

Kentaro Shimada

Faculty of Education, Soka University, 1-236 Tangi-machi, Hachioji-Shi, Tokyo 192-8577, Japan

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ABSTRACT

This study examines Kenya's Free Primary Education policy's long-term effects on intergenerational mobility (IM). Children are likely to achieve a higher level of education than their mothers and fathers, especially those whose parents had no education. Analyzing the 1985–1994 cohort identifies a 1–3 % shift in IM compared to the 1985–1989 cohort, primarily driven by the rise of parental schooling. Kenya shows significant IM; thus, the background does not solely determine one's future. Kenya has a rate of return similar to the global average. Parental education's impact seems limited, and a child's education alleviates poverty.

1. Introduction

Education enhances life satisfaction and economic prosperity (Oreopoulos and Salvanes, 2011), thus leading to better job opportunities and higher income and contributing to happiness (Killingsworth et al., 2023). Parents prioritize education for their children's successful future, with investments influenced by parental income and education levels, showing intergenerational persistence (Narayan et al., 2018).

Intergenerational persistence in the years of education perpetuates income inequality across generations (Becker, 2003; Corak, 2013). Examining effective educational policies, such as fee abolition, is crucial, especially in Sub-Saharan Africa (SSA), where poverty hinders access. However, universal primary education (UPE) policies overlook completion rate disparities, disproportionately affecting children from poorer households (Bennell, 2021a).¹

Free and compulsory education guarantees equal access to education as a fundamental human right. Rawls (Rawls, 2001) supports the allocation of resources to the disadvantaged through a policy of basic education. However, Sen (Sen, 2002) emphasizes the freedom derived from tuition waivers, enabling access to advanced education and financial stability. This study reconsiders the long-term utility of free primary education (FPE) policies, emphasizing Sen's fairness over Rawls'. Sen's fairness is based on the "capability approach," which focuses on the acquisition of entitlements by beneficiaries and regards state support in favor of valuing the economic advancement of disadvantaged individuals as important (Sen, 2002; Nussbaum, 2006).

The equitable provision of educational opportunities requires temporary but impartial support for groups in need (Oketch et al., 2010; Pritchett, 2001). Such equitable distribution involves value judgments that reflect society's perception of fairness in resource allocation (Rawls, 2001; IBE U, 2018). Rawls' theory of justice underpins liberalism but is challenged by contemporary philosophers (Nussbaum, 2006; Sandel, 1998; Sen, 1995) who question its assumptions and the treatment of marginalized groups (IBE U, 2018).

This study examines the impact of Kenya's 2003 fee abolition policy on the intergenerational mobility in education (IM) of disadvantaged populations. The IM measures social openness and poverty reduction, distinguishing between absolute and relative mobility (Narayan et al., 2018). Absolute mobility measures the standard of living of a generation in comparison to its predecessors, while relative mobility assesses a person's economic success in relation to their background. This study addresses two questions: (1) whether there has been upward mobility between generations as a result of expanded educational opportunities (absolute mobility) and (2) whether individuals from low-income backgrounds have achieved economic stability as a result of the FPE policy (relative mobility). The first analysis uses a decomposition method to distinguish between the expansion of parental education and the general expansion, while the second uses a quasi-experimental framework to estimate the returns for FPE beneficiaries compared to other groups.

This study integrates two streams of literature that have progressed independently thus far. The first pertains to IM (see (Black and

E-mail address: kshimada@soka.ac.jp.

¹ UPE refers to fee abolition and other related policy interventions at the primary education level. In Kenya, the policy to universalize primary education is called "Free Primary Education" (FPE). FPE in this study refers to the fee-abolition policies introduced in Kenya.

Devereux, 2011) for an overview), which addresses the emergence of global economic disparities across generations. Intergenerational economic inequality is typically measured by the persistence of assets between parents and children, which is indicative of limited social mobility in societies characterized by solid intergenerational persistence (Lee and Lee, 2021). Azomahou and Yitbarek (Azomahou and Yitbarek, 2016) reveal an increase in IM in nine SSA countries. Narayan et al (Narayan et al., 2018). also observe improved IM in most developing economies, except those in the SSA. Mobility tends to be lower in developing countries than in advanced economies, as evidenced by a sample of 148 economies worldwide.

The second strand of literature focuses on the returns to education. Investment in human capital is crucial for economic growth, with investment in basic education encouraged by Basic Human Needs and international cooperation to promote human development. However, Hanushek and Woessmann (Hanushek and Woessmann, 2012) warn that mere years of education do not necessarily lead to economic growth. Despite an average global rate of return of approximately 10 % (Psacharopoulos and Patrinos, 2018), the return on investment in education can be affected by various factors, such as labor market mismatches, learning biases within the education system, intra-family resource allocation, and education cost constraints (Jones et al., 2023). These studies have not adequately explored the extent to which people from low-income backgrounds break the negative cycle and build a solid economic foundation through educational attainment and subsequent life stages.

2. Literature review

2.1. Free primary education policies in Kenya and SSA

The current FPE policy in Kenya, introduced in 2003, is the third stage of efforts to abolish fees (Somerset, 2009). This policy was implemented quickly and without legal precedent or discussion following a promise made by the National Rainbow Coalition government after defeating the Kenya African National Union. The first attempt to abolish fees in the 1970s was suspended in the 1980s due to structural adjustments and austerity measures. It was later reintroduced with international support, with the Children's Rights Act 2001 and the global human rights framework Education for All at the forefront.

While the third FPE policy reduced out-of-school enrollment by approximately one million students, cost reductions were insufficient due to additional school-related expenditures and implementation challenges (Abuya et al., 2015). The subsidy provided per pupil was insufficient. It often led to delays and only partially covered the cost of instruction and materials; other necessary educational expenses were not covered (Sawamura and Sifuna, 2008). In addition, delays in administrative support for the rapidly growing number of school entrants have negatively impacted the learning environment, leading to parental concerns about the declining quality of public schools and a shift to low-cost private schools (Nishimura and Yamano, 2013).

Bennell (Bennell, 2021a) examines the factors influencing the achievement of UPE in SSA, highlighting new social class relations, the growth of higher education and private schools, and competition for jobs in the formal sector. The study emphasizes the importance of elite political commitment in achieving UPE and points out that education reforms have favored wealthier households, potentially exacerbating inequalities in access.

Although primary school completion rates in Kenya have risen from around 25–80 % since 2000, progress has stagnated since 2010. Primary school enrollment rates in rural areas have increased significantly for males, while rates for females have improved only minimally. Inequalities persist in higher education, with wealthier quintiles having disproportionately higher enrollment rates, indicating a preference for wealthier groups. Bennell (Bennell, 2021b) also examines the challenges of implementing UPE in SSA, focusing on marginalized groups such as

women, Muslims, people with disabilities, rural and nomadic students, and refugees.

While the FPE policy aims to create equal educational opportunities for all, the distribution of grants across the country raises concerns that marginalized groups are not benefiting sufficiently, thus perpetuating inequity. The Sustainable Development Goals and Kenya's revised Constitution and Basic Education Act emphasize the importance of equity (NTPSDFP, 2020).

Lucas and Mbiti (Lucas and Mbiti, 2012) report that the FPE policy improved primary school completion rates, especially among disadvantaged students. Moussa and Omoeva (Moussa and Omoeva, 2020) compare three African nations and find that fee abolition policies increased completion rates and reduced early marriages, with limited labor market impact. Ozier (Ozier, 2018) finds that the Free Day Secondary Education (FDSE) policy reduced early marriages but did not enhance cognitive skills. Duflo et al (Duflo et al., 2015). tracked nearly 20,000 individuals over seven years and showed that health education reduced fertility rates but did not fully explain the reasons when combined with financing. Brudevold-Newman (Brudevold-Newman, 2021) finds that the FDSE policy increased female education, delayed childbirth, and shifted work from agriculture to skilled jobs, highlighting the varied effects of education policies on socioeconomic factors.

2.2. Studies on absolute and relative mobility

Absolute mobility measures changes in one's economic status compared to one's parents. Yonatan (Yonatan, 2022) and Hossein (Hossein, 2022) show declines in many countries post-20th century, while Erik et al (Liss et al., 2019). note lower mobility for recent cohorts. In SSA, Christians generally exhibit better mobility than traditionalists and Muslims (Alesina et al., 2023), and IM worsens over time, affecting income inequality, growth, and poverty dynamics (Seife, 2023; Christian, 2023). Foreign aid, especially for education, promotes upward IM (Compaoré et al., 2022), while ethnicity and historical factors also play a role (Patricia, 2022). IM studies in developing nations need to distinguish between absolute and relative mobility, as the former indicates economic progress and the latter reflects income inequality, aiding in policy design.

Relative mobility focuses on how an individual's economic success is influenced by their background. Black and Devereux (Black and Devereux, 2011) and Holmlund et al (Holmlund et al., 2011). highlight the role of upbringing, parental education, and social environment in resource transfer between generations. Chevalier and Lanot (Chevalier and Lanot, 2010) stress the importance of family environment over genetic inheritance. Studies show that education returns can disrupt intergenerational persistence, with education increasing labor productivity and justifying investment (Psacharopoulos, 2004). However, empirical research in developing regions is limited, often using parental education as an instrumental variable (Kuepie et al., 2009; Soderbom et al., 2006), and regional and cultural factors also influence educational investment (Little et al., 2009), necessitating comprehensive analyses considering different contexts.

3. Methodology

3.1. Identification strategy

3.1.1. Estimation of absolute mobility

The transition probability matrix calculates discrete educational outcomes for parents (generation t) and children (generation $t+1$). p_{ij} represents the probability of educational outcomes that move from state i in t to state j in $t+1$, where $i, j = 1 \dots 4$ are four distinct educational outcomes, "no education (NE)," "primary education (P)," "secondary education (S)," and "tertiary education (T)." The upward (downward) mobility indicators were calculated as the average values of the four entries below (above) the diagonal of the child-parent education

transition matrix. The immobility ratio is calculated as the average value of the four entries on the main diagonal of the intergenerational education transition matrices (Heineck and Riphahn, 2007).

Following Cattaneo et al (Cattaneo et al., 2007), this study used a decomposition technique to ascertain the extent to which observed educational attainment was explained by parents' educational endowment and the evolving pattern of IM over time. The probability that a child belongs to cohort t observed in educational category j is decomposed as follows:

$$\Pr(y_j)_t = \sum_{j=1}^J [\Pr(y_i | y_j^p)_t \times [\Pr(y_j^p)_t]] \quad (1)$$

where y_j^p indicates parental educational attainment, with $j=1\dots 4$, denoting the alternative educational outcomes. The first term in the summation measures the conditional transition rates, and the second is the marginal distribution of parental education. The change in the proportion of individuals in j can result from a change in either the conditional $\Pr(y_i | y_j^p)_t$ or marginal distribution of parental education $\Pr(y_j^p)_t$. The change between cohorts in probability is defined as follows:

$$\Delta \Pr(y_i) \equiv \Pr(y_i)_t - \Pr(y_i)_{t-1} = \lambda + \eta \quad (2)$$

where $\lambda = \sum_{j=1}^J \Delta \Pr(y_i | y_j^p)_t \times \Pr(y_j^p)_t$ measures educational expansion independent of parental education. This term contains information about the general trend in educational attainment j and, thus, changes in mobility. The second term, $\eta = \sum_{j=1}^J \Delta \Pr(y_j^p)_{t-1} \times \Pr(y_j^p)_t$, measures the change in educational attainment j between cohorts caused by changes in parental background effects. $\Pr(y_j^p)_t$ and $\Pr(y_j^p)_{t-1}$ in λ and η , respectively, are used as weights.

The IM over time is often analyzed using a univariate regression to be estimated separately for each cohort, such as

$$c_i = \alpha + \rho f_i + \varepsilon_i \text{ for } i = 1, \dots, N. \quad (3)$$

Here, $c_i = C_i/\sigma_c$ and $f_i = F_i/\sigma_f$ are the number of years of education of child i (C_i) and the father (F_i) normalized by their corresponding standard deviations (σ_c , σ_f , respectively), ε_i is an error term, and ρ is the correlation coefficient. The advantage of estimating the correlation coefficient ρ is that it determines the difference in the variance of educational attainment across generations. The coefficient ρ could be interpreted as a measure of the inequality of opportunities due to circumstances independent of a child's effort. However, changes in ρ capture not only changes in the father-child education transmission but also different phenomena, such as the secular rise in schooling and changes in compulsory education.

3.1.2. Estimation of relative mobility

This study estimated the wage effect of an individual's schooling using a wage function. It introduced the FPE intensity index to address self-selection bias in the above schooling behaviors and the sample selection bias simultaneously (Wooldridge, 2002). This study uses the following procedure. First, T_i^* in the joint Heckman-Instrument (IV) first-stage selection estimates probit $P(D_{wi}=1 | T_i^*) = \Phi(\theta' T_i^*)$ over the entire wage worker born between 1985 and 1994 incorporates all exogenous explanatory variables to calculate the inverse mill's ratio (selectivity term). The second step is the 2SLS (Two-Stage Least Squares) using the instrument and the selectivity term. Z_i^* in the second-stage IV equation for S_i is a vector of the newly estimated λ_i (θT_i^*) and T_i^* for all observations in the selected subsamples:

$$D_w = 1[\theta' T_i^* + \delta_i > 0] \quad (4)$$

$$S_i = \theta Z_i^* + \theta X_i + \mu_i \quad (5)$$

$$\ln W_i = \beta_0 + \beta_j S_i + \beta_{ij} X_i + \lambda_i (\theta T_i^*) + \nu_i \quad (6)$$

Here, S_i is the number of years of schooling, and X_i denotes a vector of demographic covariates, current marital status, and regional fixed effects.

A combination of birth cohort and regional factors determines the FPE intensity. First, the study identifies the cohort affected by FPE and then considers the degree of regional influence within that cohort (Lucas and Mbiti, 2012; Moussa and Omoeva, 2020; Brudevold-Newman, 2021). To establish FPE benefit values, the study determines theoretical gains for students enrolled in each grade level as of 2005. For instance, a student who was eight years old in 2005 is assumed to receive a 100 % benefit because they would have been six years old, the official school age, in 2003. Similarly, a nine-year-old student is assumed to receive 7/8ths (88 %). This method allows for the estimation of theoretical benefit values based on the age at which the benefit is received.

However, a challenge arises regarding the non-exposed group depicted in Fig. 1, which is supposed to have no FPE beneficiaries. However, many are still enrolled in primary education. To address this issue, the study (1) determined enrollment figures for each age group in 2005, (2) assigned theoretical values based on grade levels enrolled, and (3) calculated the average for each age group to adjust the intensity index. This method considers the current students in each age group, dealing with overage enrollments.

Second, regional intensity is calculated using the rate of out-of-school children (OOSC) before the introduction of FPE in each district. The decrease in tuition would induce "inframarginal, credit, and capacity constraint students" [31p. 3] to prefer primary school. To confirm that a higher rate of OOSC with the introduction of FPE would attract more children, the study used the 1999 census as a reference, approximating the rate before the introduction of FPE. Using this assumption, the average rate of OOSC was then calculated for each district. The adjusted cohort intensity index and the district's average truancy rate were combined and subtracted from 1 to determine the FPE intensity index.²

3.2. Data

This study utilized data from the Kenya Integrated Household Budget Survey (KIHS) 2015–2016, comprising 24,000 household observations (KNBS, 2015–2016). The analysis focused on individuals born between 1985 and 1994, aged 21–30 years. Of the total sample of 92,781 in the survey, 13,582 (6.8 %) were born between 1985 and 1994 (aged 21–30 years). Of these, 61 % are heads of households or spouses in the variable (b03) related to the relationship with the head of household, which corresponds to 8304 individuals: 32.4 % are heads of households and 28.4 % are spouses, and the rest (about 10 %) are other relationships (e.g., grandparents/children, sister/brother, nephew/niece, and other relatives).

The sample is limited to co-residential with educational information on parents and children, possibly leading to wage selection bias. Most of the excluded sample are descendants who are financially independent from their parents. The household heads and spouses sample shows that the average wage is 5.8 % higher than the son/daughter sample. Married is 82 %, which is eight times higher than the son/daughter's sample. The average number of years of education is 8.3 years, less than the son/daughter's sample. Then, they tend to live in urban areas (31 % for the son/daughter's sample, but 47.8 % for the head/spouse's sample).

Since prior studies have also set up pairs of sons and daughters with

² Ideally, data before the introduction of FPE in 2003 would be used; however, due to unavailability, the 1999 census data were utilized. The high-intensity index reflects economically constrained populations with high numbers of out-of-school and over-aged children. Assuming wages negatively correlate with the FPE intensity index, 1 was subtracted from the index and inverted to ensure a positive regression coefficient.

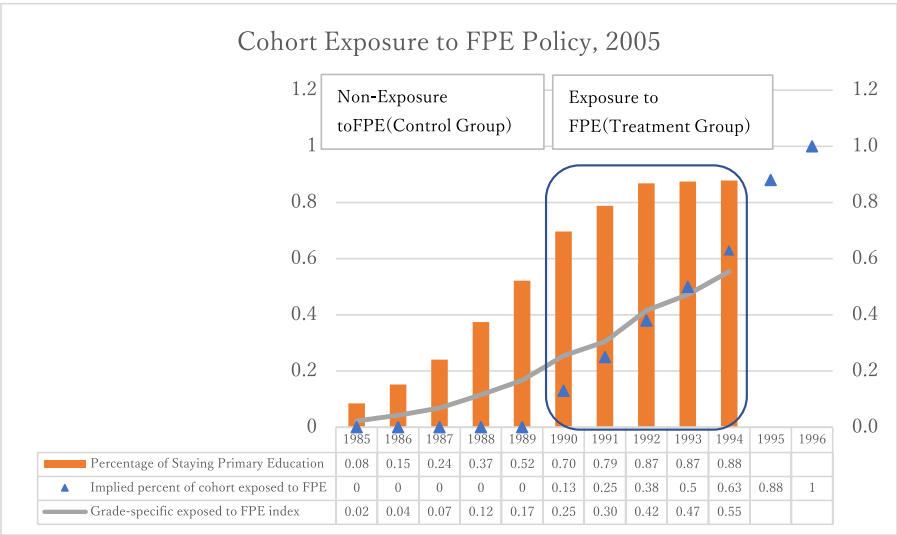


Fig. 1. Percentage of primary education enrollment rates by cohort and degree of impact of FPE. Source: Created by the author from KIHBS, 2005 (KNBS, 2005–2006). Note: The third FPE was introduced in Kenya in 2003. Thus, the eight-year-old cohort was the first to show total exposure to FPE. In 2005, the cohort from 1985–1994, aged 11–20 years, was included.

their fathers and mothers when focusing on two generations, although "grandchildren" and "grandparents" are related to the head of the household in 1.4 % and 0.4 % of the respondents, respectively, this study also focuses on parent-child pairs. The number of "sons/daughters" is 3898 (28.7 %). In the sample shown above, there are 25 cases with missing years of education, and the final number of valid observations of 3873 (2477 son-parent pairs, 1396 daughter-parent pairs) is included in the analysis of absolute mobility.

Relative mobility analyses included wage earners from the birth cohorts, with a sample size of 1463 sons/daughters. Of the 3873 child-parent pairs, 65 % do not have wage information due to the non-active status of laborers (e.g., full-time students, housewives, and retired), and 34 % of them miss information on monthly salary (d26) or "zero" hourly wage. This study estimates return to education with heads/spouse samples as a reference.

Of the 8304 head/spouse samples (62 % are female, which is nearly 100 % a spouse), 33 % are omitted due to missing information on the variables. Within the omitted samples, 98 % missed the wage information due to no information on monthly salary, working hours per week, or "zero" hourly wage.

4. Results

4.1. Absolute mobility

4.1.1. Education transition matrices

This study estimated a 4×4 transition probability matrix of educational outcomes for each father and mother-daughter cohort. These transition matrices provide information on the probability of children reaching a certain level of education, depending on their parents' educational level. For example, it enables us to understand that 20.7 % of daughters whose mothers have no education (NE) also have no education, looking at the row for mothers and the column for daughters. Similarly, 33.8 % of daughters whose mothers have no education have a transition to primary education (P), and so on.

Higher transition probabilities along the diagonal suggest a strong correlation between parental and child education levels. Children are likely to achieve a higher level of education than their mothers and fathers, especially those whose parents received no education, as 80–90 % of them completed primary or higher levels of education. However, children whose parents completed post-secondary education achieved the same level. For example, 73.6 % of daughters whose mothers had

higher education also obtained higher education, implying intergenerational persistence at the higher level of education.

4.1.2. IM, decomposition, and intergenerational correlation in education

The IM index summarizes the results above numerically (Heineck and Riphahn, 2007). The upward (downward) mobility indicator was the average of the four entries below (above) the diagonal of the educational transition matrix. The immobility rate was the average of the main diagonal elements. All indicators consider values in the interval (0,1), representing probabilities.

Table 4 shows that upward mobility is 0.41 for the mother-daughter sample and 0.40 for the father-daughter sample for the entire cohort (Table 4, Panel A). The upward mobility rate increased by 3–5 percentage points for both the mother and father-daughter samples. The sons' sample showed a similar trend, but the upward mobility was slightly weaker than for the daughters. Downward mobility remained at around 5–10 % for both samples.

The interest in the study of IM lies in how the expansion of educational opportunities has changed over time. This decomposition helps to understand the factors that contribute to changes in IM, including structural changes in society (general expansion) and the secular rise of parental education in shaping the educational outcomes of offspring (Table 5)³. The percentages in the last two columns represent the proportion of IM explained by the general expansion of education and growth of parental education, respectively. For instance, in Panel A, for individuals who attained tertiary education, the negative value of general expansion (-0.4 %) for daughters suggests a slight decline in the likelihood of attaining tertiary education compared to their mothers' generation. However, the growth of parental education (1.4 %) indicates that an increase in maternal education positively influenced daughters' likelihood of attaining tertiary education.

For sons, the positive general expansion (0.2 %) indicates a slight

³ General expansion [λ] represents the overall increase or decrease in the likelihood of attaining a specific level of education for the offspring generation compared to the parental generation.; Growth of parental education [η] captures the effect of changes in parental education levels on the likelihood of their children attaining a certain level of education; Cohort change [$\Delta Pr(y_i)$] reflects changes in the distribution of educational attainment within the population over time, unrelated to parental education. It measures shifts in educational outcomes for the younger generation compared to the older generation.

Table 1

Highest level of education completed by cohort.

	Parents (%)				Children (%)				Obs.
	NE	P	S	T	NE	P	S	T	
Parents-Son Pairs									
<i>BY1985–1994 (Age 21–30)</i>									
Mother	32.6	48.4	14.0	5.0	5.7	31.1	42.5	20.7	2477
Father	50.8	26.2	15.1	7.9	5.7	31.1	42.5	20.7	2477
<i>BY1985–1989 (Age 26–30)</i>									
Mother	34.3	48.4	12.1	5.2	7.0	40.2	33.0	19.8	758
Father	55.5	23.9	12.4	8.6	7.0	40.2	33.0	19.8	758
<i>BY1990–1994 (Age 21–25)</i>									
Mother	31.9	48.3	14.8	4.9	5.1	27.1	46.7	21.1	1719
Father	48.7	27.4	16.4	7.6	5.1	27.1	46.7	21.1	1719
Parents-Daughter Pairs									
<i>BY1985–1994 (Age 21–30)</i>									
Mother	28.4	47.4	17.8	6.5	6.8	25.1	41.8	26.4	1396
Father	49.4	24.9	16.5	9.2	6.8	25.1	41.8	26.4	1396
<i>BY1985–1989 (Age 26–30)</i>									
Mother	37.0	42.7	15.1	5.2	10.2	33.3	31.3	25.3	384
Father	55.2	24.0	14.1	6.8	10.2	33.3	31.3	25.3	384
<i>BY1990–1994 (Age 21–25)</i>									
Mother	25.1	49.1	18.8	7.0	5.5	21.9	45.8	26.8	1012
Father	47.2	25.3	17.4	10.1	5.5	21.9	45.8	26.8	1012

Note: NE stands for "no education," P for "primary," S for "secondary," T for "tertiary," and Av.Yr. for "average years of education." For instance, the category "primary education" includes a person who completed primary and attended some primary school.

Table 2

Summary statistics.

Birth Year 1985–1994 (Age 21–30)	Sons/Daughters		Heads/Spouses	
Variable	Mean	SD.	Mean	SD.
Log hourly wage	3.42	0.85	3.63	0.97
Year of education	9.46	3.84	8.97	4.14
Female	0.30	0.46	0.51	0.50
Married	0.10	0.30	0.78	0.41
Urban	0.31	0.46	0.51	0.50
Experience	9.49	4.84	11.36	4.97
Experience ²	113.39	108.67	153.84	125.85
Household size	6.36	2.59	3.50	1.83
Log per adult consumption expenditure	8.24	0.63	8.75	0.68
Mother's education	4.87	4.27		
Father's education	4.16	5.04		
Inverse Mill's ratio (selectivity term)	0.59	0.31	0.22	0.24
FPE intensity	0.62	0.18	0.70	0.16
N		1463		5549

increase in the probability of attaining tertiary education compared to their fathers' generation. The negative growth in parental education (-0.2 %) suggests that an increase in paternal education slightly affects the probability of tertiary education for sons. Panel B also shows similar interpretations, but only for those who have completed secondary school.

The decomposition analysis compares the 1990–1994 cohort with the 1985–1989 cohort. Positive differences indicate higher values for the younger generation. The cohort changes by about 1–3 % (except mothers and sons, which remain consistent at 0.1 %), which is likely due to the comparison cohorts being contiguous and covering younger generations with a short span of five years. In addition, the component explained by the increase in the parents' level of education varies from 40–88 %. For daughters who reached the higher education stage, the change in the educational level of the father and mother accounts for about 80 % of the change. For sons, the change in the cohort is explained by the increase in the educational level of the father and mother, which is around 40–50 %.

Although the changes are slight, the IM of the younger generation is higher, and there are different trends for daughters and sons. In particular, daughters' IM may be explained by the rise in the education of their parents. The analysis of the factors contributing to parents'

Table 3

Education transition matrices in birth cohort 1985–1994.

Daughter	NE	P	S	T	Total	Obs.	Total
<i>Mother</i>							
NE	20.7	33.8	31.6	13.9	100	396	28.4
P	1.7	28.6	50.2	19.5	100	661	47.3
S	0.8	10.5	41.5	47.2	100	248	17.8
T	0.0	1.1	25.3	73.6	100	91	6.5
Total	6.8	25.1	41.8	26.4	100	1396	100.0
<i>Father</i>							
NE	12.3	27.7	38.8	21.2	100	690	49.4
P	2.0	33.9	49.1	14.9	100	348	24.9
S	1.3	15.2	46.5	37.0	100	230	16.5
T	0.0	4.7	28.9	66.4	100	128	9.2
Total	6.8	25.1	41.8	26.4	100	1396	100.0
<i>Son</i>							
<i>Mother</i>							
NE	15.4	36.6	35.2	12.9	100	808	32.6
P	1.3	35.6	48.2	14.9	100	1198	48.4
S	0.3	13.0	49.0	37.8	100	347	14.0
T	0.0	2.4	17.7	79.8	100	124	5.0
Total	5.7	31.1	42.5	20.7	100	2477	100.0
<i>Father</i>							
NE	9.9	34.0	41.5	14.6	100	1258	50.8
P	2.0	40.5	44.8	12.6	100	649	26.2
S	1.1	17.6	50.9	30.4	100	375	15.1
T	0.0	6.7	25.1	68.2	100	195	7.9
Total	5.7	31.1	42.5	20.7	100	2477	100.0

Source: Kenya Integrated Household Budget Survey (KNBS, 2015–2016)

Note: The "NE" category includes those who have not attended or completed primary school. The "P" and "S" categories include those who have attained primary and secondary (lower and upper) school, respectively. The "T" category pertains to those who attained at least post-secondary schooling.

educational increase is beyond the purpose of this study.

Table 6 presents the intergenerational correlation in education, with Panels A, B, and C representing different birth year cohorts for the individuals observed. The elasticity values (β) represent the effect of parental education (mother and father separately) on the educational attainment of their sons and daughters. For instance, in Panel A, an additional year of schooling in the mother's education is associated with a 0.26-year increase in the son's education and a 0.27-year increase in the daughter's education, all else equal. Similarly, an additional year of schooling in the father's education is associated with a 0.12-year

Table 4
IM by parent and birth cohort.

		Daughter				Son			
		UM	DM	IM	Total	UM	DM	IM	Total
<i>Panel A: Whole Cohort 1985–1994</i>									
(Age 21–30)	Mother	0.33	0.07	0.41	1.00	0.31	0.06	0.45	1.00
	Father	0.31	0.09	0.40	1.00	0.30	0.09	0.42	1.00
<i>Panel B: Younger Cohort 1990–1994</i>									
(Age 21–25)	Mother	0.34	0.06	0.39	1.00	0.32	0.05	0.45	1.00
	Father	0.32	0.08	0.39	1.00	0.31	0.09	0.41	1.00
<i>Panel C: Older Cohort 1985–1989</i>									
(Age 26–30)	Mother	0.29	0.07	0.46	1.00	0.29	0.08	0.45	1.00
	Father	0.29	0.10	0.42	1.00	0.27	0.10	0.46	1.00

Source: Kenya Integrated Household Budget Survey (KNBS, 2015–2016)

Note: UP, DM, and IM represent upward mobility, downward mobility, and immobility, respectively.

Table 5
Decomposition of IM, change between 1985–1989 and 1990–1994 cohorts

Birth Year in 1985–1989 and in 1990–1994 (Age 21–30)		General Expansion	Growth of Parental Education	Cohort Change	General Expansion	Growth of Parental Education
		[λ]	[η]	[ΔPr(yi)]	%	%
<i>Panel A: Attained "Tertiary"</i>						
Mother	Daughter	−0.4 %	1.4 %	1.0 %	22.7 %	77.3 %
	Son	0.2 %	−0.2 %	0.1 %	58.7 %	41.3 %
Father	Daughter	−0.6 %	2.4 %	1.9 %	19.0 %	81.0 %
	Son	−0.7 %	−0.7 %	−1.5 %	49.3 %	50.7 %
<i>Panel B: Attained "Secondary"</i>						
Mother	Daughter	1.5 %	1.5 %	2.9 %	50.4 %	49.6 %
	Son	0.1 %	1.0 %	1.1 %	11.6 %	88.4 %
Father	Daughter	0.7 %	1.1 %	1.8 %	37.5 %	62.5 %
	Son	0.6 %	1.0 %	1.7 %	37.5 %	62.5 %

Source: Kenya Integrated Household Budget Survey (KNBS, 2015–2016)

Table 6
Intergenerational correlation in education.

	<i>Panel A Birth Year 1985–1994</i>		<i>Panel B Birth Year 1985–1989</i>		<i>Panel C Birth Year 1990–1994</i>	
Elasticity(β)	Son	Daughter	Son	Daughter	Son	Daughter
	1	2	3	4	5	6
	Eduyear	Eduyear	Eduyear	Eduyear	Eduyear	Eduyear
Mother Education	0.258*** (0.017)	0.271*** (0.021)	0.313*** (0.032)	0.241*** (0.047)	0.223*** (0.020)	0.261*** (0.024)
Father Education	0.124*** (0.013)	0.111*** (0.017)	0.146*** (0.025)	0.192*** (0.037)	0.114*** (0.015)	0.095*** (0.020)
Constant	9.106*** (0.362)	9.408*** (0.624)	8.955*** (0.455)	12.044*** (0.775)	9.358*** (0.464)	8.683*** (0.652)
Observations	2477	1396	758	384	1719	1012
R-squared	0.237	0.347	0.359	0.442	0.228	0.353
F-test	10.21	10.21	10.21	10.21	10.21	10.21
Correlation(ρ)	Son	Daughter	Son	Daughter	Son	Daughter
	7	8	9	10	11	12
	Eduyear	Eduyear	Eduyear	Eduyear	Eduyear	Eduyear
Mother Education	0.278*** (0.018)	0.292*** (0.022)	0.338*** (0.035)	0.260*** (0.050)	0.240*** (0.021)	0.282*** (0.026)
Father Education	0.151*** (0.016)	0.135*** (0.021)	0.178*** (0.031)	0.234*** (0.046)	0.139*** (0.019)	0.116*** (0.024)
Constant	2.095*** (0.083)	2.164*** (0.144)	2.060*** (0.105)	2.770*** (0.178)	2.153*** (0.107)	1.997*** (0.150)
Observations	2477	1396	758	384	1719	1012
R-squared	0.237	0.347	0.359	0.442	0.228	0.353
F-test	10.21	10.21	10.21	10.21	10.21	10.21

Robust standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1, region fixed effects are included in the specifications.

increase in the son's education and a 0.11-year increase in the daughter's education.

The coefficient ρ represents the correlation between parental education and offspring education. The features of intergenerational correlation are likely to apply to other normalized measures of mobility. The intergenerational correlation considers the variance in the educa-

tional distribution of parents and children (Emran et al., 2018). For instance, in Panel A, an additional year of schooling in the mother's education is associated with a 0.28-year increase in the son's education and a 0.30-year increase in the daughter's education, all else equal. Similarly, an additional year of schooling in the father's education is associated with a 0.15-year increase in the son's education and a

0.14-year increase in the daughter's education.

There is a notable drop in both elasticity and correlation for both mother's and father's education from the 1985–1989 cohort to the 1990–1994 cohort for sons. For daughters, while the elasticity of a mother's education increases slightly from the 1985–1989 to the 1990–1994 cohort, the correlation also increases, indicating a stronger intergenerational link over time. Compared with sons, daughters generally show a stronger correlation and elasticity with both parents' education, especially the mother's. The influence of the father's education appears more pronounced in the earlier cohort (1985–1989) for both sons and daughters.

4.2. Relative mobility

4.2.1. Economic success of the descendants

Table 7 shows that one year of education increases wages by 9.3 % for the heads/spouses and by 6.9 % for the sons/daughters. Education plays a decisive role in determining wages, as shown by the consistently positive correlation between additional years of education and higher wages in all models. However, a gender pay gap persists, with women consistently earning lower wages across all models. Marital status also impacts wage levels, with marriage generally having a positive effect on wages. Similarly, residing in urban areas, larger household sizes, and higher consumer spending tend to result in higher wages, although the significance of this effect varies depending on the model and demographic category. In addition, parents' level of education has a positive effect on their children's wages. This reflects the intergenerational transmission of human capital and underscores the importance of parental education in shaping future income potential, although the magnitude is relatively weak.

Finally, the FPE intensity index is positively associated with wages.

Table 7
Marginal Effect of Education on Wage.

Birth Year 1985–1994 VARIABLES	Head/Spouse			Son/Daughter		
	OLS 1	IV 2	IV 3	OLS 4	IV 5	IV 6
	Lwage	Lwage	eduyear	Lwage	Lwage	eduyear
Year of education	0.095*** (0.006)	0.093*** (0.006)		0.069*** (0.010)	0.069*** (0.010)	
Female	−0.348*** (0.023)	−0.283*** (0.037)	−0.035 (0.024)	−0.134*** (0.043)	−0.229*** (0.058)	−0.021 (0.035)
Married	−0.041 (0.030)	−0.023 (0.030)	−0.009 (0.019)	0.148** (0.074)	0.422*** (0.125)	0.230** (0.083)
Urban	0.073*** (0.026)	0.092*** (0.027)	−0.007 (0.017)	0.084* (0.050)	−0.012 (0.063)	−0.035 (0.037)
Working experience	−0.012 (0.009)	−0.022* (0.010)	−1.043*** (0.007)	0.016 (0.016)	0.129** (0.047)	−0.979*** (0.028)
Working experience ²	0.002*** (0.000)	0.002*** (0.000)	0.003*** (0.000)	0.001 (0.001)	−0.003 (0.001)	0.001 (0.001)
Household size	0.050*** (0.009)	0.050*** (0.009)	0.009 (0.005)	0.008 (0.010)	−0.005 (0.011)	0.000 (0.006)
Log monthly per adult total consumption expenditure	0.407*** (0.024)	0.404*** (0.024)	0.042** (0.014)	0.228*** (0.044)	0.203*** (0.045)	−0.010 (0.025)
Mother's years of education				0.003 (0.006)	0.002 (0.006)	0.001 (0.003)
Father's years of education				0.010** (0.005)	0.009* (0.005)	0.003 (0.003)
Selectivity term (inverse mill's ratio)		−0.216* (0.095)	0.0307 (0.060)		1.021* (0.407)	0.638** (0.243)
FPE_intensity			15.94*** (0.053)			14.95*** (0.087)
Constant	−0.865*** (0.236)	−0.666** (0.236)	9.241*** (0.142)	0.897** (0.449)	0.096 (0.597)	9.571*** (0.333)
Weak identification test (Cragg-Donald Wald F-statistic)			86012			29872
t-statistics (FPE)			301			172
Observations	5549	5549	5549	1463	1463	1463
R-squared	0.31	0.31		0.24	0.24	
F-statistics	7.95	41.13		7.88	8.17	

Region fixed effects are included in the specifications. Robust standard errors are in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Lwage is the log hourly wage, and eduyear is years of education (dependent variables).

This suggests that investment in primary education can positively impact wage levels, underscoring the importance of education policy in promoting economic development and improving livelihoods. The FPE intensity instrument appears sufficiently strong. Table 7 contains weak identification tests, such as the Cragg-Donald Wald F-statistic. A high F-statistic (86012 and 29872) indicates that the instrument is strong and jointly significant in predicting educational attainment, which supports its validity.

To summarize, the FPE intensity instrument is suitable for explaining variations in educational attainment. It shows relevance, strength, and significant predictive power, indicating its suitability as an instrumental variable in the wage equation. However, additional robustness and sensitivity analyses may be warranted to confirm its validity and assess possible biases.

4.2.2. Robustness check: the validity of the FPE intensity index (instrument)

Three scenarios are examined in this study to check the validity of the FPE intensity index. The first scenario applies the true FPE intensity index to a placebo sample, aged 31–40 years. The second scenario uses the fake OOSC rate as a proxy for regional intensity in 2015, the post-FPE year. The third scenario replaces the cohort intensity index with that of the fake cohort (1975–1984 cohort, 31–40 years old). The instruments can be considered relevant if the joint hypothesis of the excluded instruments yields an F-statistic greater than 10 (Moussa and Omoeva, 2020). These sensitivity checks confirm how the instruments properly function. The placebo sample, the fake OOSC rate, and the fake cohort intensity should be no or weaker than the actual specification.

In the first scenario, the placebo cohort analysis shows that the FPE intensity instrument is not valid, as indicated by a non-significant F-statistic of 0.07 (the weak identification test), probably due to the drop of the sample ($N=408$). The reduction of the sample is less co-residence

of their parents in 30–40 s decedents. This implies that the placebo cohort is inappropriate for using the FPE intensity instrument. On the other hand, the second scenario and the third scenario demonstrate that the fake regional and cohort intensities are valid instruments. However, the weak identification tests show less F-statistics. The fake instruments show F-statistics of 29720 in column 4 and 2900 in column 6, but the actual instrument shows 29872 of the F-statistics in column 6 of Table 7.

While the placebo cohorts produced the expected results, the substitutions of the regional and cohort intensity indices were not fully in line with expectations. A possible explanation of the validity of the second scenario is that the composition of OOSC among the districts between 1999 and 2015 has not changed much. The number of OOSC has decreased over the decade, but the trend of regional disparity still exists as a reflection of socioeconomically disadvantaged areas such as Arid and Semi-Arid Lands. The third scenario using the older cohort's FPE intensity shows relatively less validity. The possible explanation is that some cohorts might enjoy FPE due to repeating or late entry into primary education. These scenarios show the FPE intensity instrument is sensitive to the change of cohorts.

Overall, these analyses underscore the need to properly construct the FPE intensity index to capture intended policy effects. It is crucial to interpret the results of the manipulated variables as differences in the wage effect of education across groups and to exercise caution in attributing the effects solely to the FPE policy.

5. Discussion

5.1. Absolute mobility

Children are likely to achieve a higher level of education than their mothers and fathers, especially children with parents with no education. For children of "no-education" parents, 80–90 % have reached primary or higher levels of education. However, this trend is different for the more educated. Children whose parents are highly educated reach the same level. The 1985–1994 generation affected by the FPE policy experienced a 1–3 % change in IM compared to those born between 1985 and 1989. This change was largely driven by the secular rise of parental schooling, with variations ranging from 44 % to 88 % depending on parent-child combinations. The elasticity and correlation of IM range from 0.25 to 0.29 for the entire cohort. It is worth noting that these results are based on coresident pairs of parents and children, indicating a 20–30 % downward bias with the non-coresident sample (Dendir, 2023). Longer-term comparisons or different demographic groups could lead to different results (Alesina et al., 2023; Dendir, 2023; Razzu and Wambile, 2022).

In Greece, a 30 % increase in advanced-level education and a 26 % increase in middle-level education were observed over 30 years, from the 1940s to the 1970s (Daouli et al., 2010). The IM shows little difference between younger and older generations, suggesting similar outcomes for those affected and unaffected by FPE policies. With less than 10 % experiencing downward mobility and around 30 % upward mobility, Kenya exhibits notable IM, indicating that one's background does not solely determine one's future. Unlike earnings, which fluctuate

Table 8
Robustness Check: Different Specifications of Instruments.

Son/Daughters	Placebo Cohorts (Born in 1975–1984)		Actual Cohorts (Born in 1985–1994)			
	Actual FPE Intensity (OOSC rate in 1999 / 1985–94 FPE exposure)		Fake Region Intensity (OOSC rate in 2015)		Fake Cohort Intensity (1975–84 FPE exposure)	
	IV 1	2	IV 3	4	IV 5	6
VARIABLES	Lnhwage	Eduyear	Lnhwage	Eduyear	Lnhwage	Eduyear
Year of education	–1.081 (4.575)		0.069*** (0.010)		0.074*** (0.013)	
Female	0.713 (3.897)	0.843*** (0.239)	–0.229*** (0.058)	–0.021 (0.035)	–0.228*** (0.058)	–0.163 (0.106)
Married	0.714 (1.966)	0.430 (0.326)	0.422*** (0.125)	0.230** (0.083)	0.414** (0.126)	1.083*** (0.240)
Urban	0.578 (1.346)	0.296 (0.263)	–0.012 (0.063)	–0.035 (0.037)	–0.011 (0.063)	–0.178 (0.111)
Working experience	–0.826 (2.688)	–0.590*** (0.129)	0.129** (0.047)	–0.979*** (0.028)	0.129** (0.047)	–0.520*** (0.081)
Working experience ²	0.006 (0.008)	0.002 (0.003)	–0.003 (0.001)	0.001 (0.001)	–0.003 (0.002)	–0.007** (0.002)
Household size	0.068 (0.239)	0.051 (0.040)	–0.005 (0.011)	0.000 (0.006)	–0.004 (0.011)	–0.060*** (0.018)
Log monthly per adult total consumption expenditure	1.545 (5.043)	1.092*** (0.238)	0.203*** (0.045)	–0.010 (0.025)	0.200*** (0.045)	0.210** (0.078)
Mother's years of education	0.058 (0.247)	0.054 (0.030)	0.002 (0.006)	0.001 (0.003)	0.002 (0.006)	0.023* (0.010)
Father's years of education	0.003 (0.051)	0.008 (0.027)	0.009* (0.005)	0.003 (0.003)	0.009 (0.005)	0.029*** (0.008)
Selectivity term (inverse Mill's ratio)	0.181 (0.846)	–0.065 (0.645)	0.069*** (0.010)	0.638** (0.243)		3.001*** (0.718)
FPE_intensity		–1.645 (6.403)		14.95*** (0.087)		371.4*** (6.898)
Constant		11.254 (6.590)	0.096 (0.597)	8.307*** (0.333)	0.071 (0.594)	–306.87*** (5.852)
Weak identification test (Cragg-Donald Wald F-statistic)		0.07		29720		2900
t-statistics (FPE)		–0.26		172.4		53.9
Observations	408	408	1463	1463	1463	1463
R-squared	0.49		0.24		0.24	
F-statistics	1.01		8.17		7.95	

Region fixed effects are included in the specifications except for placebo cohorts. Placebo cohorts prove that some county dummies and FPE intensity were dropped for collinearities if the equation has region fixed effects. The result shown here is the equation without region fixed effects. Robust standard errors are in parentheses, *** p<0.01, ** p<0.05, * p<0.1.

due to lifestyle changes, educational qualifications remain stable. Therefore, educational mobility helps mitigate biases in analyzing persistent transitory shocks and life cycles (Mazumder, 2005; Piraino, 2015).

Overall, the results of the intergenerational correlation underline the significant influence of parental education on children's educational attainment, with stronger associations observed for maternal education than for paternal education. The results remain consistent across different birth cohorts, which indicates the stability of intergenerational correlations in education. IM in South Africa shows similar trends (Kwenda et al., 2015). IM ranges from 0.27 to 0.34 in the 1984–1993 birth cohort. Furthermore, the IM has increased by more than 50 % over the past four decades. The relatively greater increase in IM among younger cohorts may be due to government commitments or support from international society to promote education for all and invest in the education sector. Education can change the balance of power between households. Better-educated mothers may spend more money on direct and efficient child development activities (Black and Devereux, 2011).

5.2. Relative mobility

The return to education analysis offers insights into how one's educational attainment contributes to breaking the poverty cycle. The global average marginal return to education is approximately 10 %, indicating consistency with global evidence (Psacharopoulos and Patrinos, 2018). Given that global estimates encompass surveys from developed societies, the relatively higher marginal return to education observed in Kenya is comprehensible. Specifically, the marginal return to education for sons/daughters is 6.9 %, whereas for heads/spouses, it is 9.3 %. Kimenyi et al (Kimenyi et al., 2006). show that the rate of return was 11 % for males and 5.7 % for females. However, this study's results (compared to the sample of sons and daughters) show a 10–30 % decrease in the rate of return over a few decades. The country-contextual economic structure creates systematically different labor markets across sectors of the economy (Jones et al., 2023).

The decline in the rate of return may have various causes. First, coresident parent-child pairs cause downward biases due to unmarried individuals earning less than non-coresident heads/spouses (Emran et al., 2018). Second, the dominance of Kenya's informal sector and the increase in the number of higher education graduates may have led to a relative decline in the education premium, which is a sign of over-education (Sam, 2018).

The influence of parental education on wages appears to be limited, with paternal education having an effect of only 1 %. Even though the economic benefit of education may be relatively lower for people who are heads or spouses, there remains a noticeable advantage to being educated. Personal educational attainment contributes to poverty reduction. More importantly, earnings only underestimate intangible and non-monetary benefits such as an increase in self-esteem, health improvements, social networks created in school, and social benefits (Moussa and Omoeva, 2020; Dickson and Harmon, 2011; Cuartas, 2022).

Viewing FPE policy changes as natural experiments, this study performs IV estimation and finds a 10-percentage point decrease in wages for heads of families/spouses, with a similar effect observed for sons/daughters. The IV estimates of returns on education in the literature typically exceed those obtained from OLS by at least 20 %. Card (Card, 1999) interprets that the return on education varied across a population, and the treatment effect worked for a subgroup called the "local average treatment effect." The decrease in tuition would induce these infra-marginal, credit, and capacity-constrained students to prefer primary school, thus explaining the relatively lower earnings compared to household heads/spouses. Furthermore, IV estimates of the return to education in Nigeria show no statistical significance (Uwaifo Oyelere, 2010).

Despite some evidence of inconsistencies in the curriculum or

insufficient acquisition of knowledge and skills through school attendance alone (Bennell, 2021a), previous research shows that UPE in Uganda has decreased delayed enrollments and increased grade completion, especially among girls in poor households (Nishimura et al., 2008). It is also clear that education significantly reduces the early marriage of girls (Brudevold-Newman, 2021) and improves wages, though not fully.

The study has several limitations. First, the exclusive focus on households where individuals live with their parents may overlook individuals with independent livelihoods who are more likely to be married and have higher income potential.⁴ In addition, an overaged group in the sample suggests a concentration of high school and college students in the 20–30 age group, which affects the interpretation of the results. Second, the study could not accurately identify the beneficiaries of the FPE policy, which limits the ability to directly attribute the results to the effects of this policy. However, the results suggest that education can aid socioeconomically disadvantaged individuals in overcoming poverty, albeit indirectly. Finally, the study does not consider the broader externalities of education, such as its impact on health or long-term IM between different age groups. Considering these factors could lead to a more comprehensive understanding of the potential of education to alleviate poverty.

6. Conclusion

This study examined the long-term effects of FPE on reducing inequality. Children are likely to achieve a higher level of education than their mothers and fathers, especially children with parents with no education. The elasticity and correlation of IM range from 0.25 to 0.29 for the entire cohort. The study examines the IM for the 1985–1994 cohorts and shows a change of 1–3 % compared to the 1985–1989 cohort. Although parental education contributes to this change, there are differences. IM in Kenya is evident: less than 10 % are affected by downward mobility and about 30 % by upward mobility, emphasizing that origin alone does not determine the future. Parental education influences the educational attainment of children across all birth cohorts, indicating stable correlations between generations.

The analysis of the return on education provides information on how the level of education influences career advancement. At 6.9 % for sons/daughters and 9.3 % for heads of families/spouses, Kenya has a rate of return similar to the global average. Despite the co-resident sample bias downward, education still provides economic benefits. An individual's educational attainment especially outweighs parental influence, reducing poverty. The study uses the FPE policy changes as natural experiments to identify a 10 % wage decline. Despite some shortcomings in educational outcomes, education plays a role in improving wages, highlighting its contribution to poverty reduction. Therefore, addressing educational problems is crucial for promoting socioeconomic improvements.

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CRediT authorship contribution statement

Kentaro Shimada: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

⁴ Emran et al. [51] find that co-residential bias in intergenerational regression coefficients and correlations was downward, indicating greater mobility. The bias was more severe in coefficients than in correlations, with correlations showing about a third of the bias and less sensitivity to co-residency rates.

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